

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2025

Course: Btech

Branch : CSE

Semester : VI

Subject Code & Name: Embedded Systems(BTCOE604C _Y23)

Max Marks: 60

Date:19/01/2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	RT-level design mainly describes		1				
	<table border="1"> <tr> <td>a. Physical layout</td> <td>b. Gate-level connections</td> <td>c. Register transfers and operations</td> <td>d. Machine code</td> </tr> </table>	a. Physical layout	b. Gate-level connections	c. Register transfers and operations	d. Machine code		
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2	Custom processor optimization aims to improve		1				
	<table border="1"> <tr> <td>a. Code portability</td> <td>b. Performance and power</td> <td>c. User interface</td> <td>d. Network speed</td> </tr> </table>	a. Code portability	b. Performance and power	c. User interface	d. Network speed		
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3	A general-purpose processor is designed to		1				
	<table border="1"> <tr> <td>a. Perform one fixed task</td> <td>b. Run multiple applications</td> <td>c. Replace ASICs</td> <td>d. Avoid software</td> </tr> </table>	a. Perform one fixed task	b. Run multiple applications	c. Replace ASICs	d. Avoid software		
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4	ARM7TDMI-S supports which instruction sets?		1				
	<table border="1"> <tr> <td>a. ARM only</td> <td>b. Thumb only</td> <td>c. ARM and Thumb</td> <td>d. Java</td> </tr> </table>	a. ARM only	b. Thumb only	c. ARM and Thumb	d. Java		
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5	Embedded software development environment includes		1				
	<table border="1"> <tr> <td>a. Text editor only</td> <td>b. Compiler, debugger, linker</td> <td>c. Power supply</td> <td>d. PCB tools</td> </tr> </table>	a. Text editor only	b. Compiler, debugger, linker	c. Power supply	d. PCB tools		
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6	Power control block helps in		1				

	a. Instruction execution	b. Reducing power consumption	c. Increasing memory	d. Debugging	
7	APB divider is used to				1
	a. Divide memory	b. Control peripheral clock	c. Increase CPU speed	d. Generate interrupts	
8	SPI is generally faster than I2C because				1
	a. Uses addressing	b. Uses clock and separate lines	c. Uses acknowledgment	d. Uses arbitration	
9	Microprocessor-based systems require				1
	a. No external components	b. External memory and I/O	c. No software	d. No clock	
10	Non-functional requirements include				1
	a. Image capture	b. Performance and power	c. Button press	d. Data storage	
11	UART block diagram includes				1
	a. Shift registers and baud generator	b. ALU and cache	c. PLL and GPIO	d. ADC and DAC	
12	Which of the following is a key design challenge in embedded systems?				1
	a. Unlimited memory	b. Deterministic timing	c. Excessive user interaction	d. High storage capacity	
Q. 2	Solve the following.				12
A)	Discuss the major design challenges involved in embedded system design.				6

B)	What is a custom single-purpose processor? Explain its advantages and limitations.	6
Q.3	Solve the following.	12
A)	What is RT-level design? Explain RT-level custom processor design.	6
B)	What is an Application Specific Instruction Set Processor (ASIP)? Explain its advantages.	6
Q. 4	Solve Any Two of the following.	12
A)	Explain the basic architecture of a general-purpose processor.	6
B)	Explain pipeline operation in ARM7TDMI-S processor.	6
C)	Explain memory mapping control in embedded systems.	6
Q.5	Solve Any Two of the following.	12
A)	Explain the architecture of TIMER/COUNTER modules.	6
B)	What is auto-baud detection? Explain its operation.	6
C)	Draw and explain the SPI block diagram.	6
Q. 6	Solve Any Two of the following.	12
A)	Explain the concept of Real-Time Operating Systems (RTOS).	6
B)	Compare microprocessor-based and microcontroller-based systems.	6
C)	Explain the importance of system design and simulation.	6
*** End ***		